

Cartilage and Bone Histology

Laboratory Manual

B. Puder, Ph.D.

References:

Gartner & Hiatt, *Color Textbook of Histology*, Ch. 7 pp. 131-156

Gartner & Hiatt, *Color Atlas of Histology*, Ch. 4 pp. 65- 88

Objectives:

Upon completion of this laboratory you should be able to:

1. Identify the following types of cartilages and where in the body they may be found:
 - A. Hyaline cartilage
 - B. Elastic cartilage
 - C. Fibrous cartilage (fibrocartilage)
2. Describe the cells and the matrix in cartilage tissues and provide details about the subcellular content and the matrix content.
3. Describe the 2 processes of cartilage growth. Be able to identify each type in a digital image.
 - A. Interstitial growth
 - B. Appositional growth
4. Identify using digital images the steps within the following processes:
 - A. Intramembranous bone formation
 - B. Endochondral bone formation
5. Identify the following cell types and describe their origin and main functions:
 - A. Osteoblasts
 - B. Osteocytes
 - C. Osteoclasts

Introduction:

Dense connective tissues include:

- dense irregular connective tissue
- tendons
- ligaments
- cartilage
- bone

The function of these connective tissues is to bear large loads and these tissues are fairly noneleastic.

These tissues consist of cells which produce a matrix.

These tissues are very sturdy and metabolically active. Cartilage and bone can grow and remodel while bearing loads. Tendons, ligaments and bone can repair itself, while cartilage regenerates poorly.

Bone uses hydroxyapatite as a supporting substance and also stores calcium for the body. Regulating serum calcium is controlled by bone cells called osteoblasts and osteoclasts. The hormones calcitonin and parathyroid hormone regulate the osteoblasts and osteoclasts, which in turn, is regulated by the serum calcium concentration.

Cartilage

Specialized connective tissue

Composed of:

Cells = chondrocytes and these cells live in cavities = lacunae

Extra cellular matrix = free and fixed glycosaminoglycans (GAGs), proteoglycans, collagen

Unique qualities/properties of collagen:

No direct nerve, blood, or lymph supply

Chondrocytes have a low metabolic rate

High tensile strength

Permits reversible compression

Function:

Support

Friction free surface at articulations (joints)

Participates in load distribution

Precursor for development and growth of bones

Perichondrium = covering found on some cartilage

dense irregular connective tissue

Interface between cartilage and neighboring tissue

Highly vascular and provides nutrition to chondrocytes via diffusion

2 layers exist = outer fibrous layer and inner cellular layer

Cells in the inner layer of perichondrium can differentiate into chondrocytes

Growth of cartilage

There are 2 mechanisms for the growth of cartilage:

1. Interstitial growth = mitotic division of chondrocytes to form isogenous groups
2. Appositional growth = perichondrial cells proliferate and differentiate into chondrocytes

*** In adults, new cartilage forms only after injury

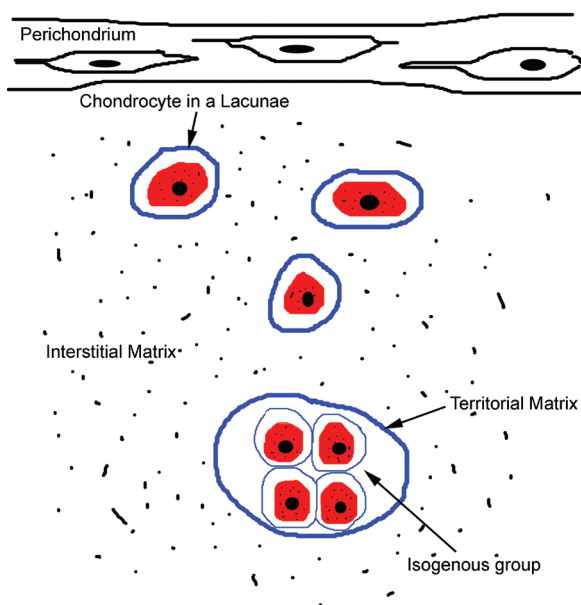


Image depicts cartilage covered by perichondrium, chondrocytes in lacuna, and interstitial growth of cartilage via mitotic cell division of the chondrocytes which then form isogenous groups.

Types of cartilage

1. Hyaline cartilage
2. Elastic cartilage
3. Fibrocartilage

Hyaline cartilage

Found:

articular surfaces of bones
walls of respiratory passages
anterior ends of ribs
epiphyseal plates of long bones

Appearance:

mature hyaline cartilage appears glassy, translucent, bluish/white

Composed of:

Chondrocytes in lacunae

Type II collagen fibrils (high tensile strength)

Matrix proteins

hyaluronic acid (free GAG)

aggrecan (proteoglycan) - maintains the hydration of the cartilage

aggrecan consists of:

protein core

side chains of:

chondroitin sulfate (GAG)

keratan sulfate ((GAG)

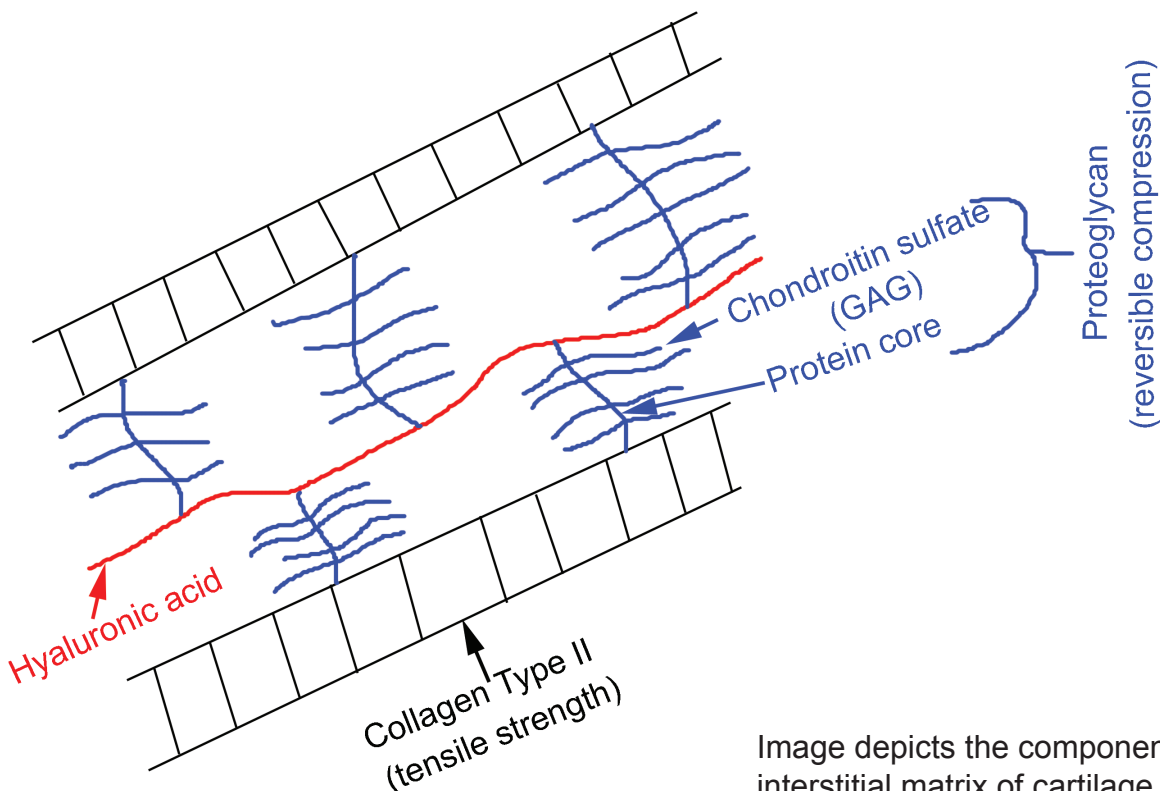


Image depicts the components of the interstitial matrix of cartilage

Development of hyaline cartilage

Mesenchymal cells

Chondroblasts

secrete matrix

chondrocytes are in lacunae surrounded by the territorial matrix
(chondrocyte = chondroblast that has become surrounded by matrix)

Chondrocytes can then grow via interstitial growth by mitotic cell division and form isogenous groups, which will push away from each other as the cells produce matrix.

Mesenchymal cells at periphery form fibroblasts which make dense irregular connective tissue called perichondrium. The cells of the inner layer of perichondrium can proliferate and differentiate into chondrocytes to create cartilage via appositional growth.

Slides 63 and 64 = Images of Hyaline cartilage

Slide 63 = trachea and esophagus

Slide 64 = lungs and bronchi

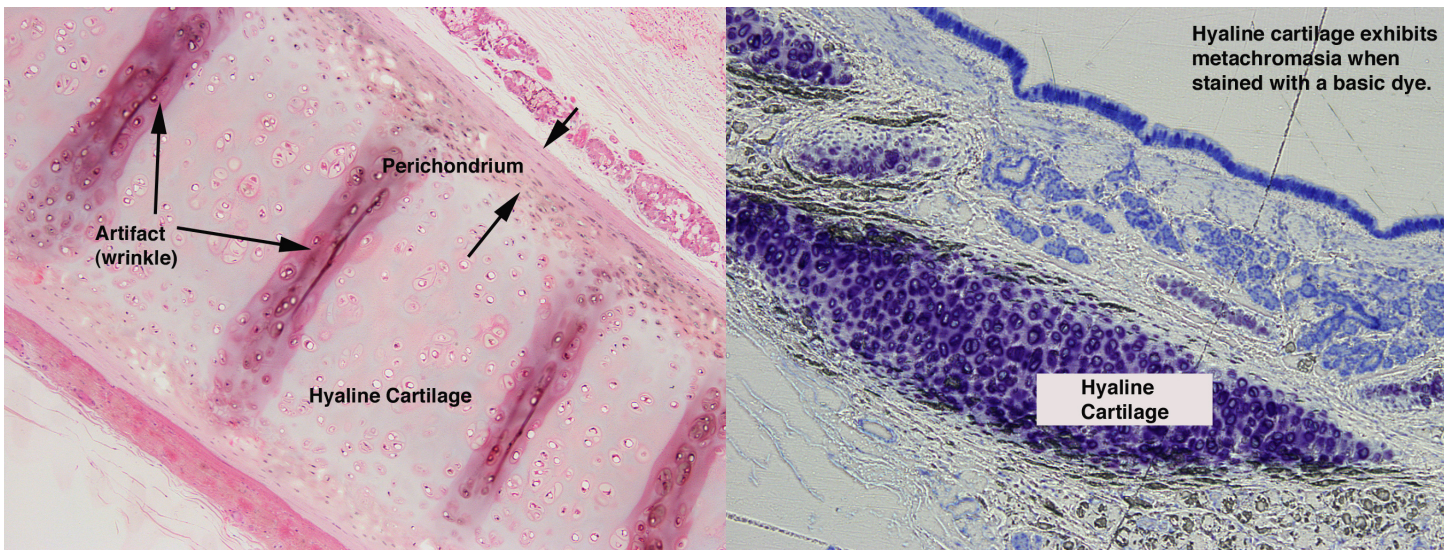


Image on the left:

The surface of **hyaline cartilage** is covered by perichondrium. When the chondroblasts within the perichondrium produce new cartilage matrix, this is called **appositional growth**.

Interior to the perichondrium are chondroblasts that produce an extracellular matrix and are then called **chondrocytes**. The space between the **chondrocyte** and the extracellular matrix is called a **lacunae**.

When these chondrocytes grow it is via mitotic cell division and is called **interstitial growth**.

The cell division of the chondrocytes creates **isogenous groups**. The newest matrix surrounding the isogenous groups stains more intensely and is called **territorial matrix**, while the older matrix is called **interterritorial matrix**.

Image on the right:

This image depicts the discontinuous plates of hyaline cartilage that surround the bronchi. The ground substance (containing proteoglycans and glycosaminoglycans) stains metachromatically.

2. Elastic cartilage

Found in regions requiring flexible support and reversible compression

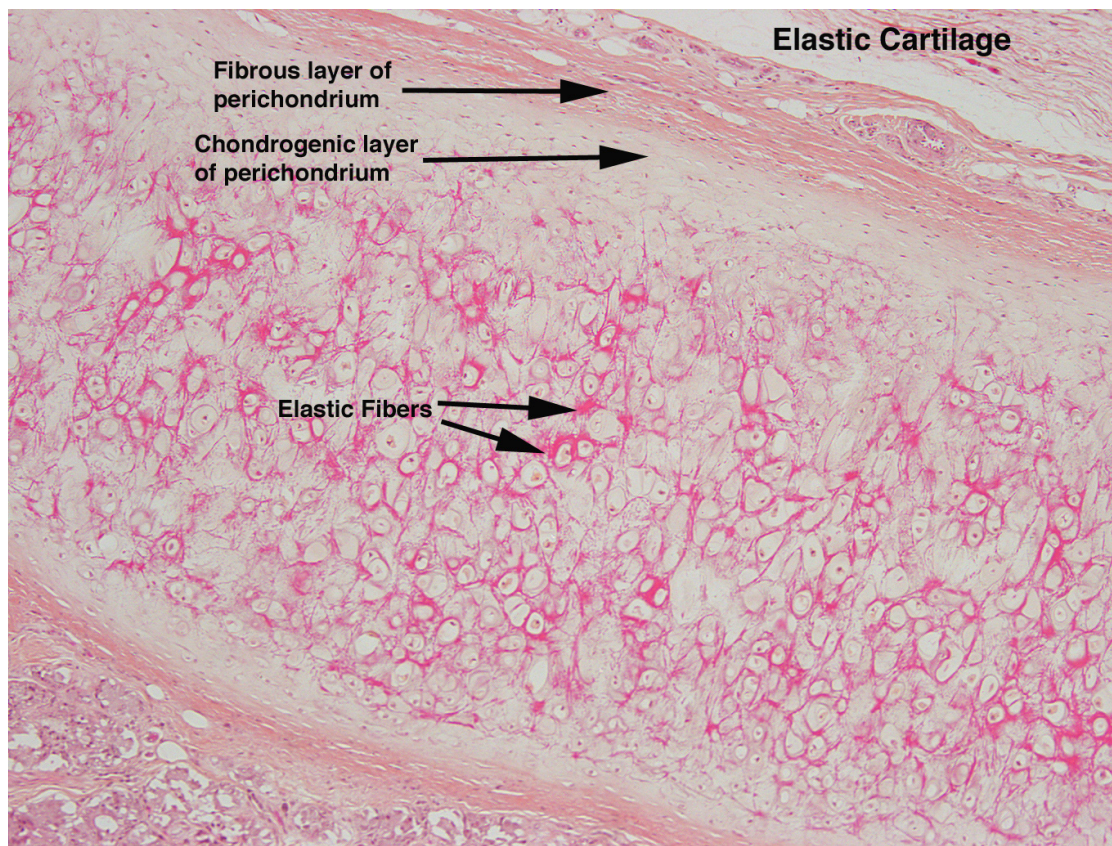
- auricle
- eustachian tube
- epiglottis
- laryngeal cartilages

Mature elastic cartilage is in circular isogenous groups with abundant elastic fibers

Development of elastic cartilage

- Mesenchymal fibroblasts produce microfibrils
- Cells differentiate into chondroblasts
- Amorphous elastin is deposited
- Perichondrium forms at peripheral edge of elastic cartilage

Image 60 Elastic cartilage (epiglottis)



This image depicts the **elastic cartilage** present in the epiglottis. This image is stained to depict the **elastic fibers** (intense pink stain).

Where are the elastic fibers located in relation to the **chondrocytes** and **lacunae**?

Observe that elastic cartilage has a well defined **perichondrium** in which one can differentiate the **fibrous perichondrial layer** from the **chondrogenic perichondrial layer**.

3. Fibrocartilage

Found:

Regions needing firm support and tensile strength
intervertebral disks
pubic symphysis
attachments of tendons and ligaments to bone

Fibrocartilage is a transitional tissue that merges tendons, ligaments, or hyaline cartilage

Features:

NO perichondrium
linear isogenous groups

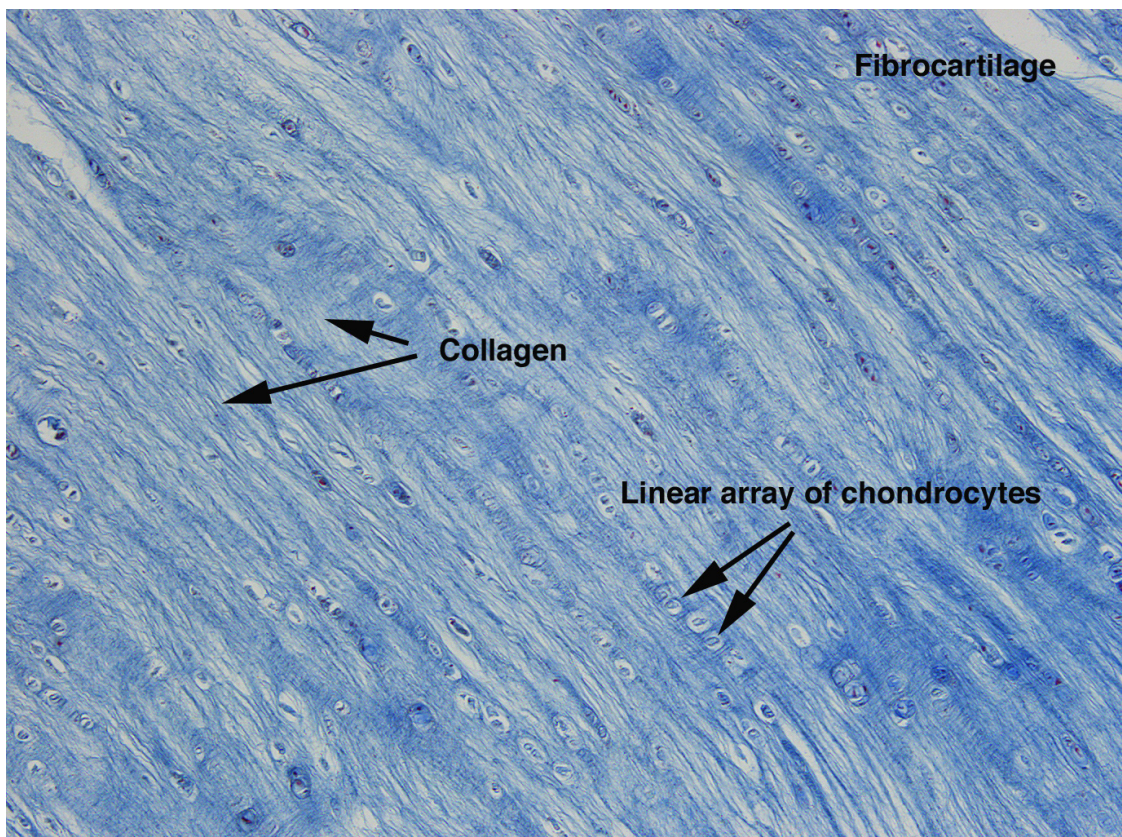
Development:

Mesenchymal cells differentiate into

chondroblasts

secrete matrix with type I collagen

Image 13 (Horse) **Fibrocartilage**



This image depicts **fibrocartilage** which is often located between hyaline cartilage and dense regular connective tissue and exhibits features from both tissue types.

Observe the **chondrocytes** surrounded by **lacunae** are arranged in **linear groups** or chains. Abundant amounts of collagen are present in the extracellular matrix.

There is no perichondrium present in fibrocartilage, so this tissue cannot grow via appositional growth. So what mechanism does it use to grow???

Cartilage types Overview

	Perichondrium	Collagen type	Elastic fibers	Isogenous groups	Occurs alone
Hyaline cartilage	Yes***	II	No	Circular	Yes
Elastic cartilage	Yes	II	Yes	Circular	Yes
Fibro-cartilage	No	I	No	Linear	No

*** except articular cartilage

Bone

Specialized connective tissue with a calcified matrix

Matrix consists of:

65% Calcium phosphate salts

35% Type I collagen, proteoglycans, glycoproteins

Cells = osteocytes which reside in lacunae embedded in matrix
(osteocytes do not divide)

Properties

Highly vascular

Nutrition via cell to cell contact via canaliculi

Growth is by apposition only

Strong in tension, but weak in torsion (bone has some flexibility)

Function

Provide support/protection

Storage space for Calcium phosphate

Transmits force of muscle contraction for locomotion

Bone cells

Osteoprogenitor cells

mesenchymal stem cells

monocyte/macrophage line

osteoblasts

osteoclasts

Bone cell types

Osteoblast

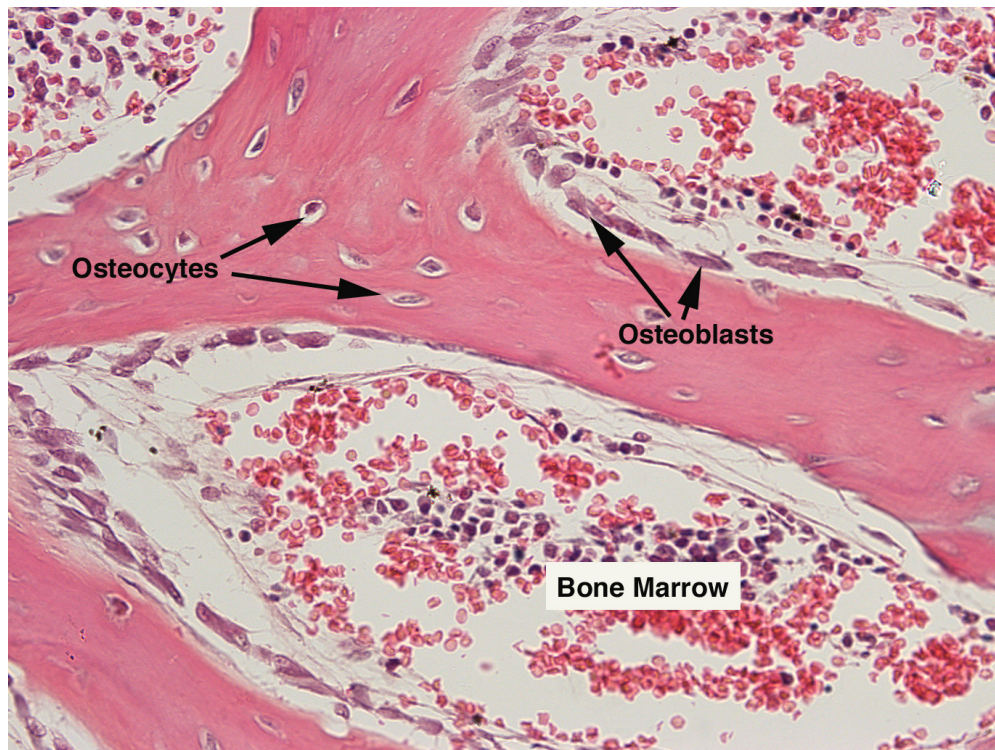
large cell, basophilic cytoplasm (increased mRNA coding for production of type I collagen)
secreted unmineralized bone matrix called osteoid (procollagen, matrix proteins, growth factors)
Responds to growth hormone and vitamin D to increase activity

Osteocyte

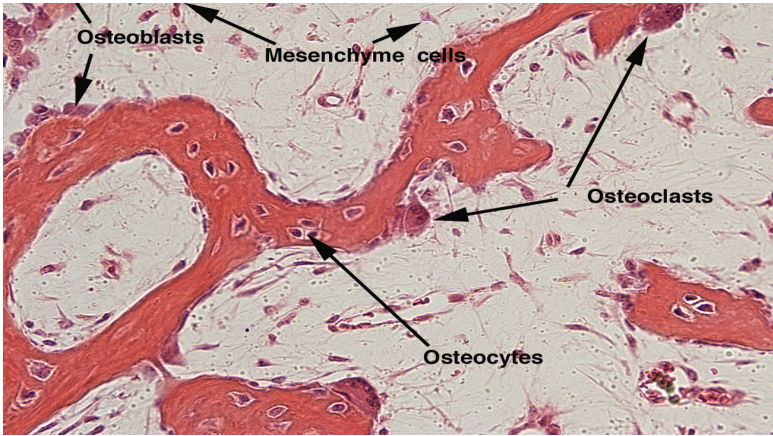
Osteoblast trapped in matrix
Is the mature bone cell in a lacuna
low synthetic activity
amitotic
has gap junctions at ends of cellular processes that provide for flow of ions and small molecules between osteocytes
receives nutrition via diffusion

Osteoclast

large, multinucleated cell
resides in a lacuna called "Howship's" lacuna (half moon shaped)
resorbs bone
has a ruffled border when active
stains for acid phosphatase (a lysosomal enzyme)
Increases activity in response to parathyroid hormone
Decreases activity in response to calcitonin



Observe the **osteoblasts** (basophilic stain) on the perimeter of the bone.
Observe the **osteocytes** in lacunae within the bone.



This image is a cross section through the developing head. Spicules of bone can be seen as the intensely stained salmon/pink structure.

Osteoblasts are present on the periphery of the spicules and are laying down bone directly, which is an example of intramembranous ossification.

Osteoclasts are large multinucleated cells also present on the periphery of the bone spicule.

Osteoclasts resorb bone so are usually located on the opposite side of the spicule as the osteoblasts.

Osteocytes are present within the bone spicule. It is an osteoblast that has been trapped within the bone matrix and cannot undergo cell division.

Macroscopic Regions of Bone

1. Trabecular (cancellous/spongy bone)
 - lattice of fine strips of bone
 - surrounds cavities filled with marrow
2. Cortical (compact bone)
 - more solid
 - contains only 10% organic material

The femur is 80% compact bone and 20% trabecular bone

Do not get regions of bone confused with developmental stages of bone:

Immature bones = primary or woven bones

Mature bones = secondary or lamellar bones

Bone lining layers

1. Periosteum
 - external lining of cells over bone
 - consists of 2 layers
 - outer layer = fibrous connective tissue with fibroblasts and collagen
 - inner layer = osteogenic layer
 - Is absent where tendons and ligaments insert on bones and at articular surfaces
 - Function:
 - supplies osteogenic cells for bone growth
 - nutrition for osteocytes

Sharpey's fibers = bundles of collagen that anchor periosteum to bone

2. Endosteum
 - single layer of cells that lines all inner surfaces of bone
 - contains osteoblasts and osteoprogenitor cells
 - Function:
 - Nutrition
 - Bone growth
 - Repair

Microscopic structure of bone

1. Cortical region (compact bone)

80-90% calcified

distinct lacuna

osteocytes are in layers = lamellae

canaliculi (projections between osteocytes) are numerous

Bone tissue and cells types are arranged in 3 different types of lamellae

A. Concentric (Haversian) lamellae

B. Circumferential lamellae

C. Interstitial lamellae

A. Concentric lamellae = Haversian system = osteon

cylindrical unit of compact bone

central part of the Haversian canal contains:

blood vessels

nerves

lymphatics

concentric lamellae of bone surround the canal

osteocytes are in lacunae between lamellae

B. Circumferential lamellae

2 types of circumferential lamellae exist:

1. outer circumferential lamellae

deep to periosteum, circularly arranged around outside of bone

2. inner circumferential lamellae

located superficial to endosteum, around new marrow cavity

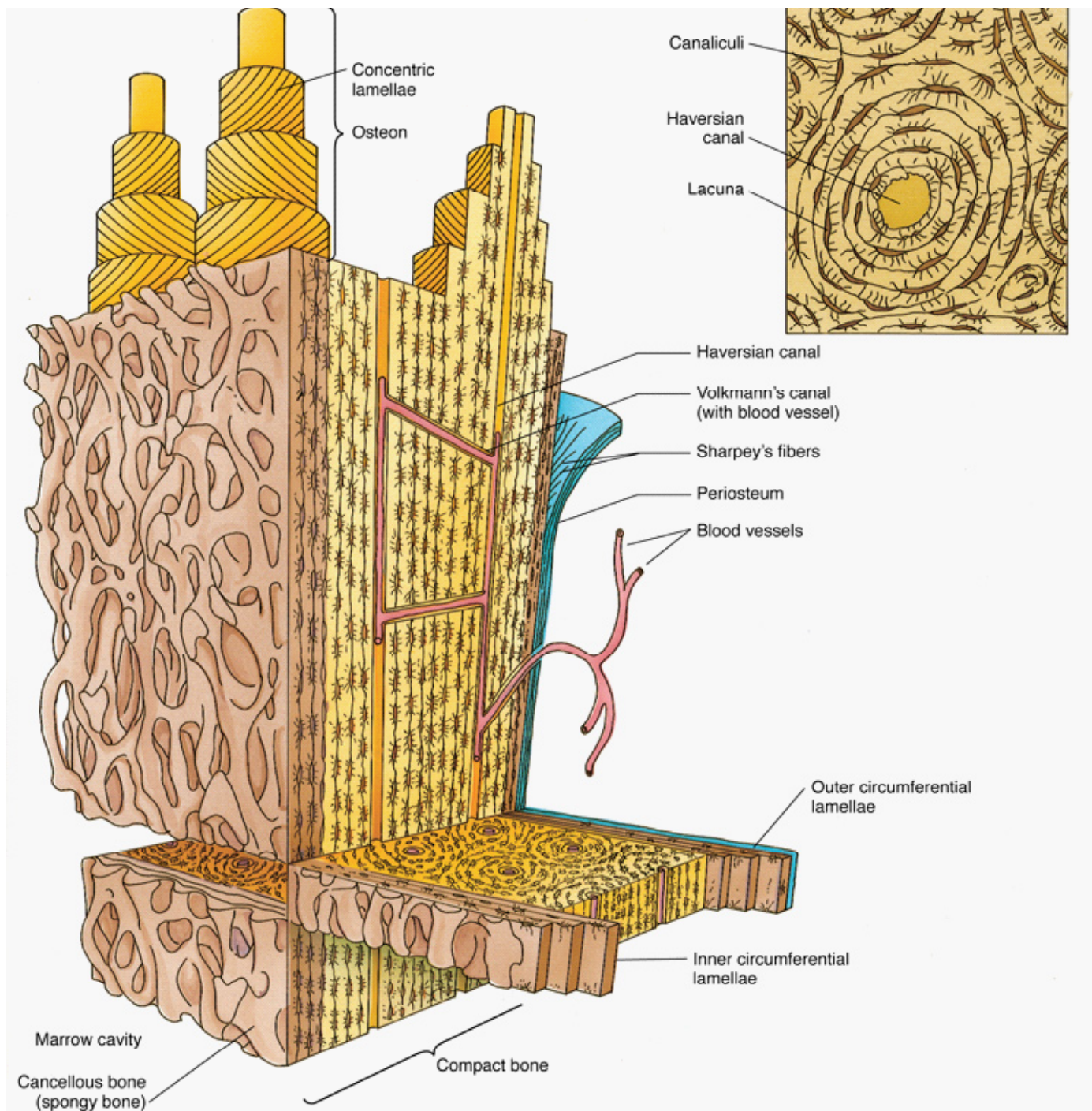
C. Interstitial lamellae

partially eroded osteon

indicates bone remodeling

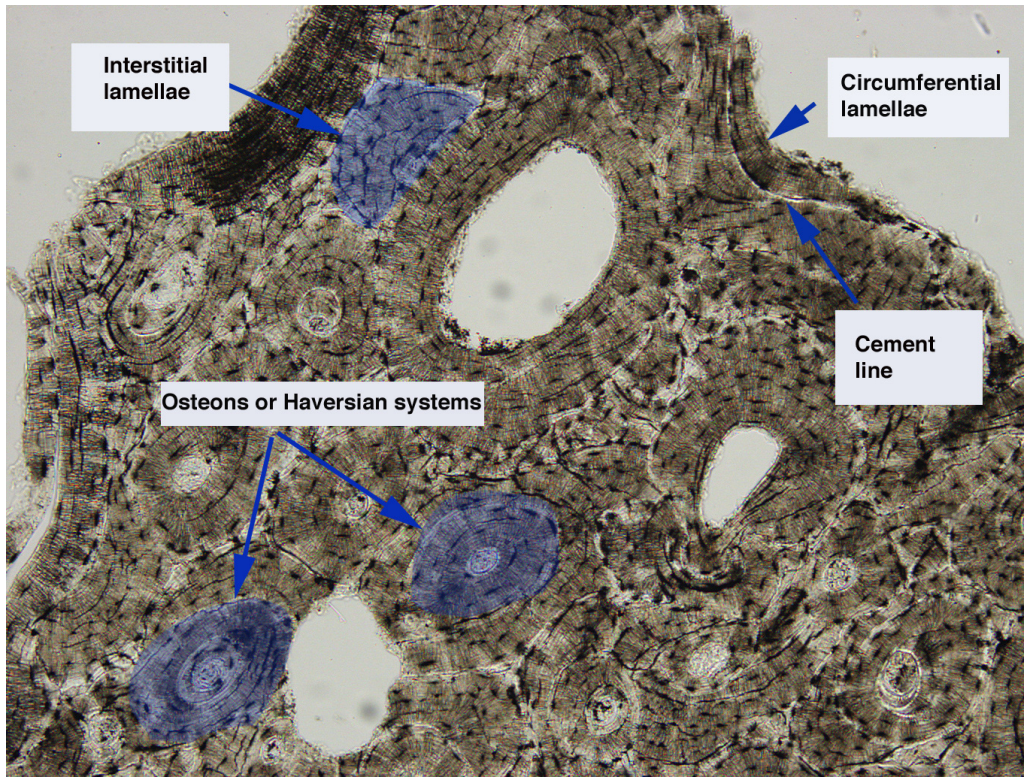
Volkman's canal = vascular connection between osteons

Cement lines = dense line where 2 osteon meet, contain mineral and collagen



Gartner & Hiatt illustration depicting compact bone, osteons, Volkmann's and Haversian canals, and cancellous (spongy) bone.

Compact bone (femur)

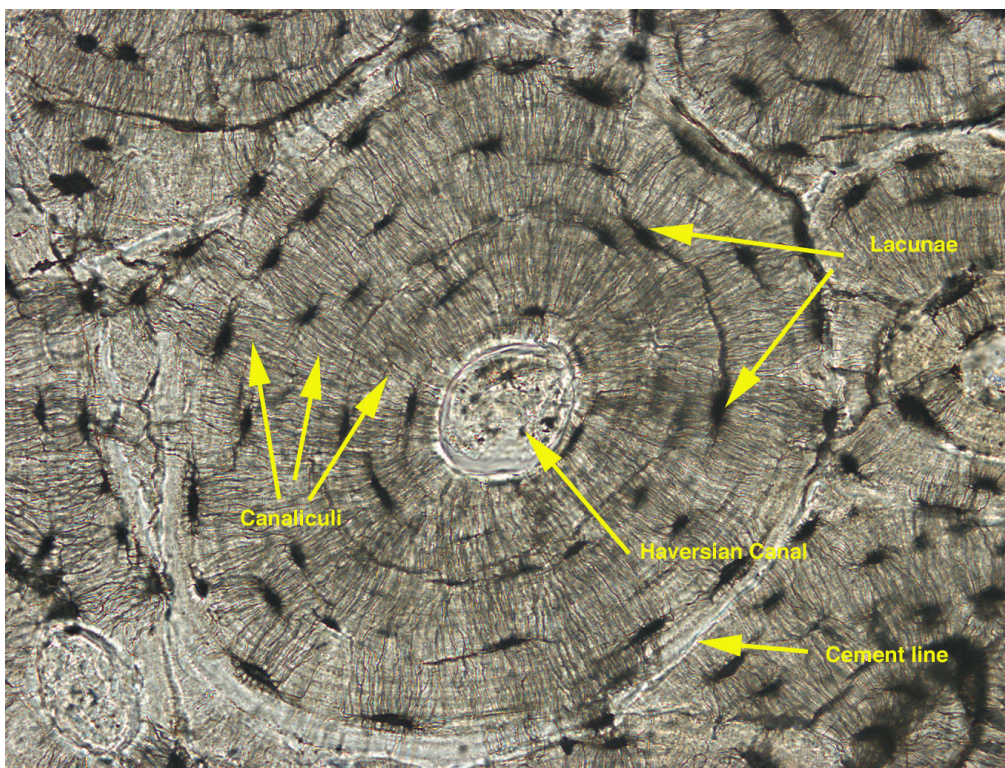


This image is a section of ground bone that has been injected with ink which helps to depict the 3 types of lamellae:

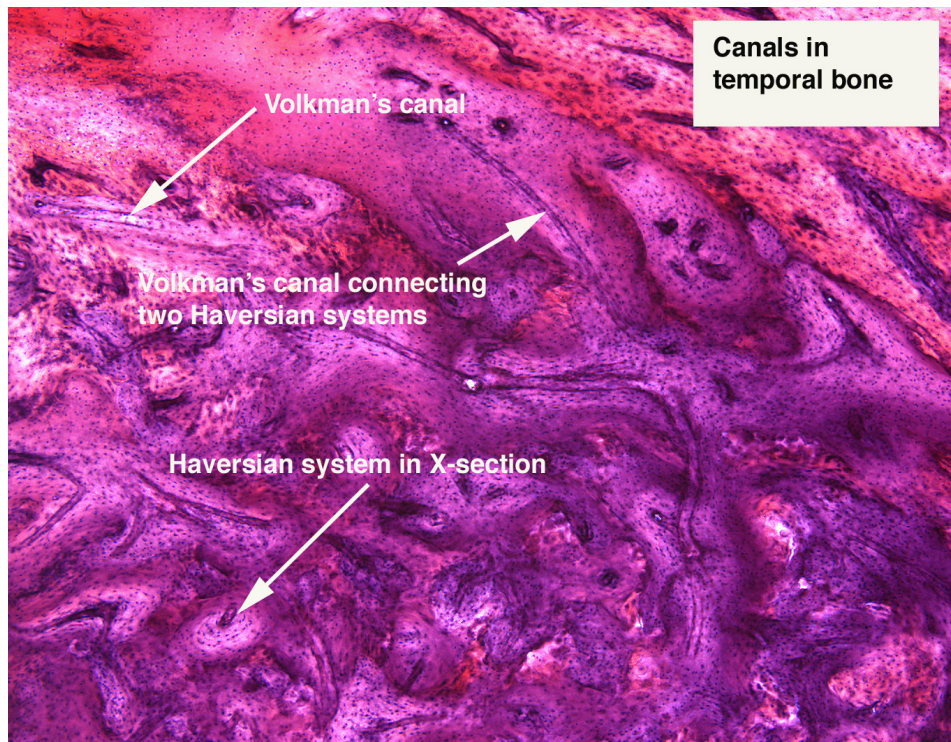
osteonic or Haversian lamellae - located within bone

surface or outer circumferential lamellae - located in the outside region of bone, but inside the periosteum

interstitial lamellae - located within bone and indicate regions of bone remodeling



This image depicts the **Haversian system**, also called an **osteon**. The **osteon** consists of the **Haversian lamellae**, **haversian canal**, **lacunae**, and **canaliculi**. Also identify the **cement lines** that are peripheral to the **osteon**.



This image is a stained section of inner ear, which depicts **Volkmann's canals**. **Volkmann's canals** run perpendicular to the long bone axis and carries vascular communication between osteons.

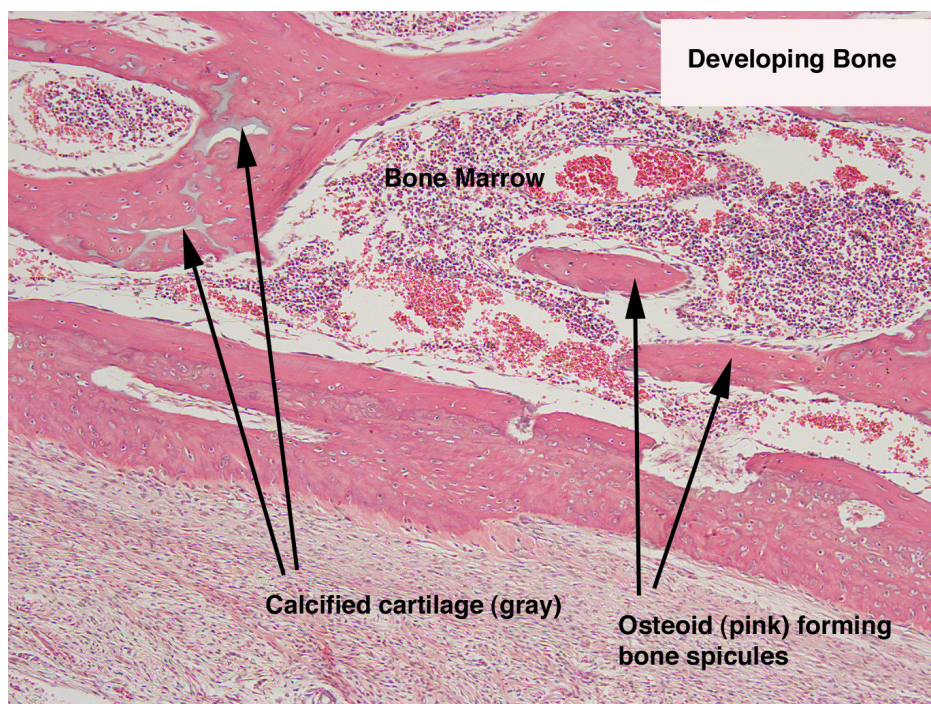
Microscopic structure of bone continued

2. Trabecular bone (cancellous bone)

Thin branching struts of bone that surround marrow

Found at cores of long bones, short bones and diploe of skull

Has similar microanatomy to cortical bone, but osteons and lamellae are less obvious



This image depicts developing bone. Note the gray areas depict cartilage that has been calcified and the pink areas depict the forming bone spicules.

Composition of Bone

A. Organic matrix (ground substance)

1. Type I collagen

Collagen is arranged linearly with collagen fibrils crosslinked

Hydroxyapatite (mineral) crystals are deposited between collagen fibers

2. Other organic components

glycoproteins

proteoglycans

B. Inorganic matrix

75% of bone is mineral, and 50% of that mineral = hydroxyapatite

Bone remodeling

Adult bone development balances bone resorption as bone is remodeled to meet stresses placed on it.

Bone Structure

Bones are classified by shape

diaphysis = bone shaft

epiphysis = bone heads

Long bones = diaphysis and 2 epiphyses

Short bones = bones have same width and length (eg. carpal bones in wrist)

Flat bones = plate-like bones (eg. skull)

Irregular bones = bones that do not fit any of the above criteria (eg. sphenoid
ethmoid bones)

Sesamoid bones = bones that develop within a tendon

Histophysiology of Bone

99% of body's calcium is stored in the skeleton

Calcium is required for many cellular activities throughout many organ systems (neurotransmission, muscle contraction, blood coagulation, cell adhesion, membrane permeability)

Calcium is released from the bones via 2 mechanisms:

1. Ionic transport of hydroxyapatite to tissue fluid (passive diffusion)
2. Stimulation of osteoclast activity indirectly through osteoblast and parathyroid hormone (PTH)

Hormonal Influences on Bone

Calcitonin (from the thyroid) inhibits the release of calcium from bone.

Growth Hormone (somatotropin from anterior pituitary) acts on the epiphyseal cartilaginous plates to increase long bone length.

Testosterone/Estrogen (gonadotropic hormones) influence the onset of ossification centers and the closure of the epiphyseal plate.

Clinical Aspects

Calcium deficiency = incomplete calcification of bone matrix and its eventual resorption
= rickets in children, osteomalacia in adults

Vitamin Problems

Lack of Vitamin D = poor absorption of Calcium from intestine which results in rickets/osteomalacia

Vitamin C - needed for normal collagen synthesis

lack of vitamin C = scurvy which causes soft bones and loose teeth

Excess Vitamin A = slows cartilage growth, accelerates ossification and early closure of epiphyseal plates

Lack of Vitamin A = decreased collagen synthesis and slows bone growth

Growth of Bone

Bones start out as cartilage and then are transformed into bone

Growth = increase in cells or matrix components which = increase in dimension of bone

Growth of cartilage as tissue

2 Mechanisms for growth

1. Interstitial (chondrocytes divide making isogenous groups)
2. Appositional (perichondrium)

Growth of Bone

There is no interstitial growth in bone, i.e. osteocytes do not divide.

Osteoblasts differentiate and proliferate to produce new extracellular matrix to increase thickness.

Osteogenesis (bone growth)

2 methods

1. Intramembranous
2. Endochondral

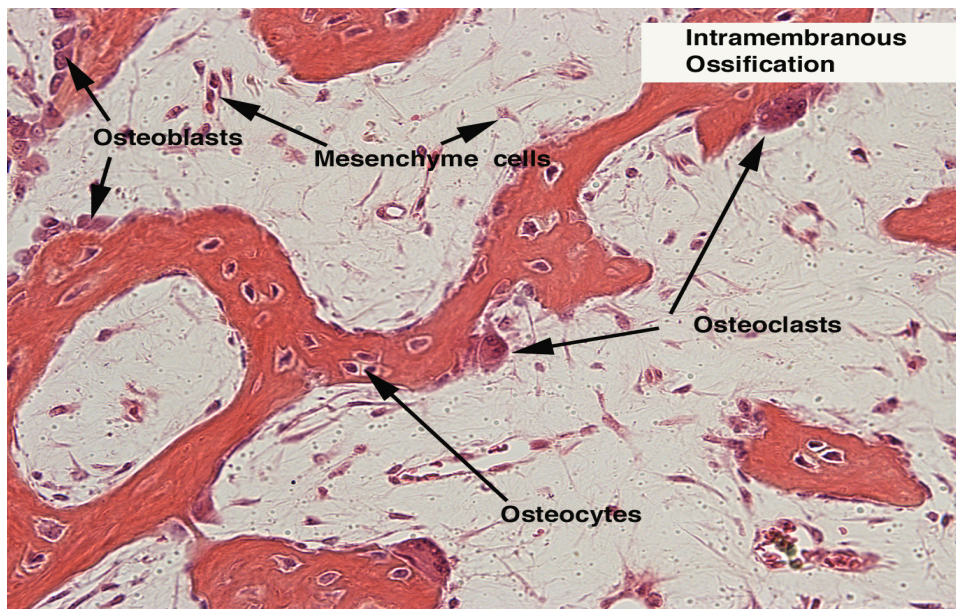
Intramembranous Ossification

Direct ossification = bone formation from mesenchyme without a cartilage precursor

Occurs in flat bones of skull, mandible

Steps in intramembranous ossification

1. Mesenchyme differentiates into osteoblasts
2. Osteoblasts form osteoid
3. Osteoid calcifies and forms bone spicules
4. Spicules join to form patches of bone (woven bone).
5. Blood vessels invade area.
6. Woven bone becomes mature lamellar bone via remodeling of osteoblasts and osteoclasts and forms layers of compact bone that surrounds trabecular bone.



This image is at the diaphysis of bone. Note the **osteoblasts**, **osteocytes**, and **osteoclasts** (residing in **Howship's lacunae**).

Endochondral Ossification

Replacement of hyaline cartilage by bone formation in the long bones and short tubular bones.

Steps in Endochondral ossification

1. Mesenchyme differentiates into chondroblast.
(this is the hyaline cartilage model in which the cartilage can grow appositionally and interstitially)
2. Chondrocytes hypertrophy.
And at the same time, perichondrium at the mid diaphysis becomes vascularized and chondrogenic cells become osteoprogenitor cells, (so perichondrium becomes periosteum) and osteoprogenitor cells become osteoblasts.
3. Osteoblasts lay down extracellular matrix and a bone collar is formed thus cutting off blood supply to the hypertrophic chondrocytes, so they die and then there are large lacunae with no cells inside them. These holes create the framework for the marrow.
4. Osteoclasts etch holes in the bone collar so osteoprogenitor and hemopoietic cells and blood vessels can enter the old cartilage model.
5. Osteoprogenitor cells divide to form osteoblasts which will put bone matrix onto calcified cartilage.

(Histo note: Calcified cartilage stains basophilic while calcified bone stains acidophilic)

6. Bone grows thick and long.

Secondary centers of ossification are at epiphysis and no bone collar is formed

Bone growth in length

Chondrocytes at the epiphyseal plate proliferate and participate in the process of endochondral bone formation.

5 zones can be seen at the epiphyseal plate

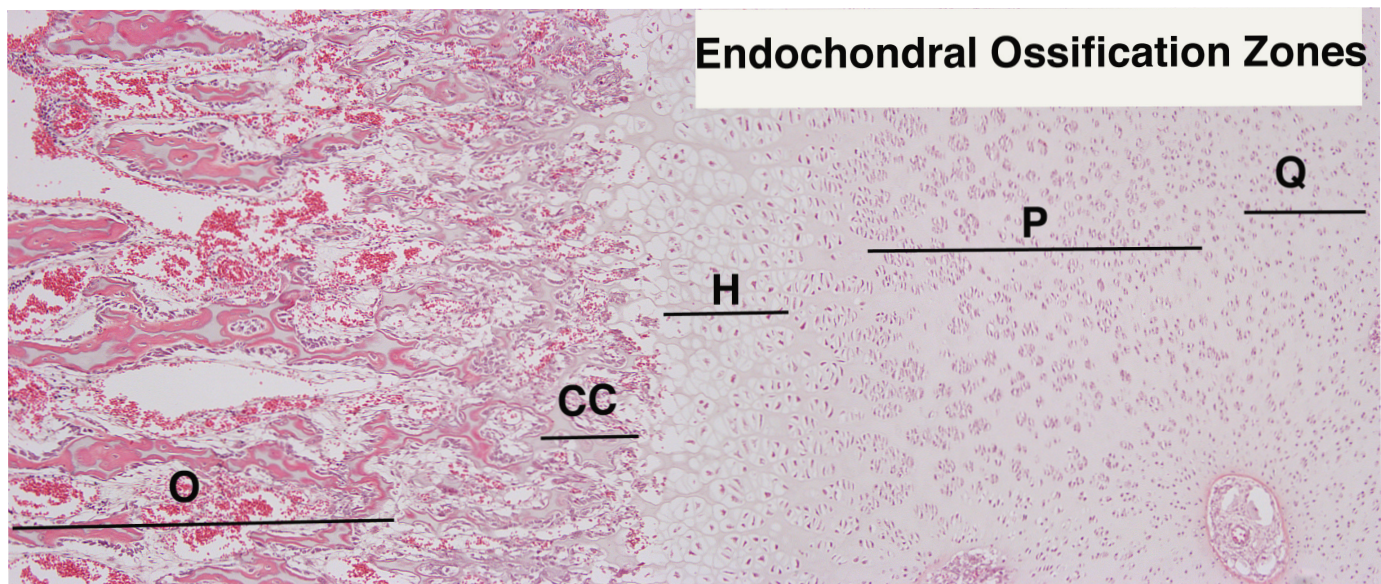
1. Zone of reserve cartilage
Chondrocytes mitotically active
2. Zone of proliferation
Isogenous groups
3. Zone of maturation and hypertrophy
chondrocytes mature, hypertrophy
4. Zone of calcification
confluent lacunae
chondrocytes die
cartilage matrix calcifies
5. Zone of ossification
Osteoprogenitor cells invade
differentiate into osteoblasts and put matrix over the calcified cartilage

If the zone proliferation rate = resorption rate then the epiphyseal plate has some width and bone grows longer. At about 20 years of age, the proliferation rate decreases so the rate of growth in the zone of proliferation is less than the rate in the zone of ossification so all the cartilage in the epiphyseal plate is replaced by calcified cartilage and then bone. The marrow cavities of the diaphysis and epiphysis become one.

Bone growth in width

Appositional growth

osteoprogenitor cells of the osteogenic layer of the periosteum proliferate and differentiate into osteoblasts. Osteoclasts must be present to resorb bone to keep the marrow space.



This image is taken at the epiphysis of the bone and depicts bone growth via endochondral ossification. Identify the 5 regions of bone development indicated in this image:

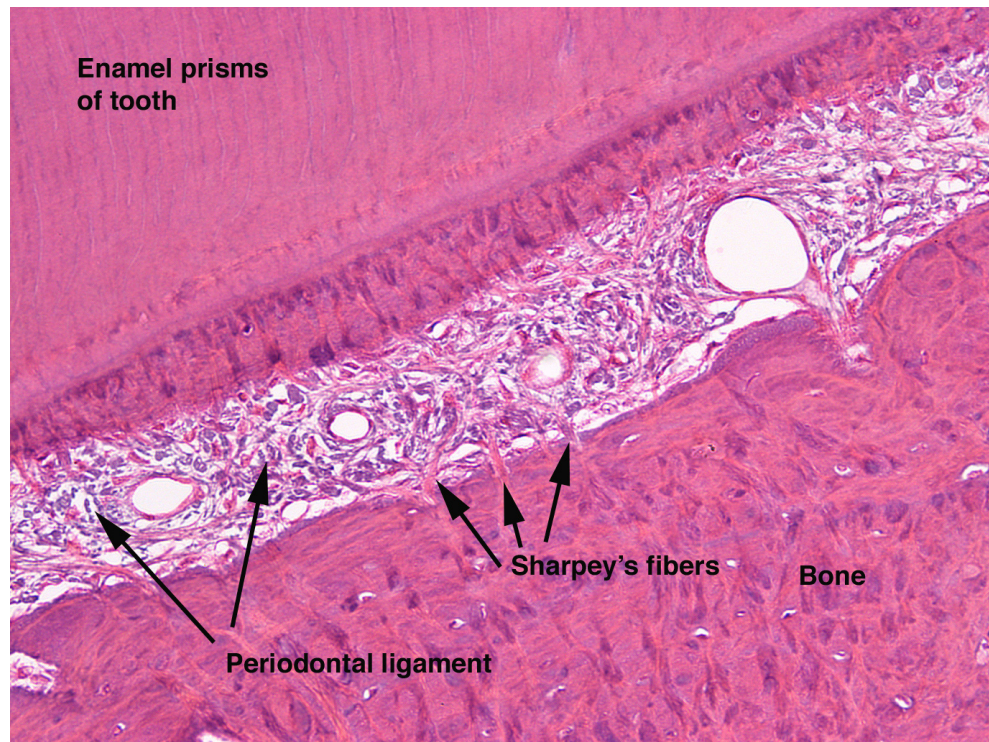
Q = Quiescent stage, or Zone of resting cartilage

P = Zone of proliferation - isogenous groups of chondrocytes are present

H = Zone of hypertrophy - chondrocytes mature and enlarge

CC = Zone of calcified cartilage = chondrocytes die and cartilage matrix calcified, lacunae may be empty

O = Zone of ossification - osteoprogenitor cells invade, differentiate into osteoblasts and lay down matrix over calcified cartilage



This image is the canine mandible and depict **Sharpey's fibers**. **Sharpey's fibers** are collagen type I fibers that are part of the periodontal ligament and penetrate the compact bone. **Sharpey's fibers** anchor the periosteum to the underlying bone.